

0.1 RQ4-Specific Coding Codebook

Coding purpose. RQ4 examines how the included reviews conceptualize building occupants, which performance metrics they make visible, whether they register imbalance between technical and human-centered metrics, and which human-centered gaps they signal.

What the scale measures. The shared 0–3 scale records treatment intensity, meaning how prominently a review treats a coded framing, metric, or gap within its own account of the literature. It does not measure how well the primary studies represent occupants, and it does not measure whether a metric is well defined or well used. A high score indicates that a review makes a topic visible and organizes part of its synthesis around it. The score describes the review, not the field.

Table 1: Shared treatment-intensity scale as applied in RQ4.

Score	Label	General interpretation
0	Not addressed	The framing, metric, or gap is not identifiable in the review.
1	Mentioned	It is named or appears indirectly, for example only as a term inside a reward formulation, without further development.
2	Discussed	It is substantively discussed as an objective, concern, or application aspect, but does not organize part of the synthesis.
3	Structuring role	It organizes part of the review’s synthesis, through a taxonomy, an explicit comparative criterion, a dedicated section, or a recurring interpretive theme.

Structure of RQ4 coding. RQ4 records one descriptive field and four survey-level scores, three of which have an associated unit-level table.

Scope class is a descriptive field naming the review’s declared focus, for example broad building control, HVAC, home energy management, occupant-centered, or real-building deployment. It is accompanied by a confirmation score recording that the review is indeed framed around the named scope. This score does not vary across the corpus and is not used in any comparison. It is retained for audit transparency only.

Occupant-framing visibility records how prominently the review treats occupants overall. Its unit table has six framings: occupant as disturbance or uncertainty; occupant as comfort constraint; occupant as feedback or preference source; occupant as explicit behavioral model; personalization or individual differences; and health, indoor environmental quality, or wellbeing framing.

Metric visibility records how prominently the review treats performance metrics overall. Its unit table has eight metrics: energy or efficiency; cost or economic objective; demand response or grid objective; thermal comfort; indoor air quality, indoor environmental quality, or health; preference or satisfaction feedback; behavioral realism or occupant interaction; and personalization or individual differences.

Metric-imbalance visibility records how far the review makes visible that the literature is more developed around technical metrics than around human-centered ones. It is coded at survey level only and has no unit table.

Gap-signal visibility records how far the review signals unresolved human-centered gaps. Its unit table has five gaps: personalization or individual differences; feedback-rich human interaction; health or indoor environmental quality beyond thermal comfort; behavioral realism or non-stationarity; and real-world human-centric validation.

Relationship between survey-level and unit-level scores. The survey-level scores for occupant-framing visibility and metric visibility are coded directly from the review and are not

derived from their unit scores. They record how prominently the review treats occupants or metrics as a whole, which is not the same as whether any single framing or metric reaches a structuring role. A review can classify one metric in detail while metrics do not organize the review overall.

Gap-signal visibility is coded directly in the same way. Across the coded corpus it coincides with the highest gap unit score in every case, which is reported here as an observed property rather than as an aggregation rule.

Shared unit. Personalization or individual differences appears as a unit on both the occupant-framing and metric-visibility tables. It is coded once and reported on both, so the two columns carry the same value for every review. Any aggregate computed across both tables should count it once.

Zero and applicability. A score of 0 means the review does not address the framing, metric, or gap. The scale does not separate a topic that a review considered and did not develop from one that fell outside its declared scope. Both are recorded as 0. This keeps denominators uniform and is consistent with the construct, which records the treatment a review gives a topic rather than the reasons for that treatment. A score of 0 should therefore not be read as evidence that a review overlooked something it could have covered. Scope class is reported alongside each review so that readers can see where a 0 follows from a narrow declared focus.

All coded reviews are scored on every dimension and unit. RQ4 has no applicability route, so the denominator is the full coded corpus.

Table 2: Operational criteria for occupant-framing visibility, applied at both survey and unit level.

Score	Criterion
0	The framing is not identifiable in the review.
1	Occupants appear only indirectly, for example as a comfort term inside a reward function or an optimization objective, with no treatment as behavioral agents, feedback sources, or personalization targets.
2	Occupants are substantively discussed through comfort, environmental adaptation, or feedback, but occupant representation does not structure the review.
3	Occupant representation organizes part of the review, appearing across several routes such as presence, activity, comfort feedback, state information, state-transition influence, and reward-related disturbance.

Operational criteria: occupant-framing visibility.

Table 3: Operational criteria for metric visibility, applied at both survey and unit level.

Score	Criterion
0	The metric is not identifiable in the review.
1	The metric is named or appears only as one term among others, without development.
2	The metric is discussed as an objective, reward component, or application concern, but does not structure the review.
3	The metric is used as an explicit comparative criterion, classified across reviewed studies, or identified as a principal objective organizing the synthesis.

Operational criteria: metric visibility.

Table 4: Operational criteria for the survey-level metric-imbalance score.

Score	Criterion
0	The relative weighting of technical and human-centered metrics is not visible.
1	An uneven emphasis is noted in passing.
2	The relative weighting is visible from how the review is organized or discussed, but the review is not mainly organized around metric imbalance.
3	Metric imbalance organizes part of the review, for example by showing explicitly that the literature is more developed around some metrics than others.

Operational criteria: metric-imbalance visibility.

Table 5: Operational criteria for gap-signal visibility, applied at both survey and unit level.

Score	Criterion
0	The gap is not signaled.
1	The gap is noted as a concern without elaboration, or is raised only in general terms such as limited real-world application.
2	The gap is explained, with its consequence for human-centered control made clear.
3	The gap is a structuring concern, developed as a recurring theme, a dedicated discussion, or an explicit basis for the review’s recommendations.

Operational criteria: gap-signal visibility.

Evidence rules. Tables, figures, taxonomies, and dedicated sections are coding evidence, and organizing use of a framing, metric, or gap in any of them supports a structuring score. Appearance of a metric only as a term inside a reward formulation supports a score of 1. Repeated mentions do not accumulate into a higher score. Content appearing in the abstract or keywords alone is scored at most 1. Supplementary material was not coded, because its availability is inconsistent across the corpus and coding it would make reviews non-comparable. Coding reflects the review authors’ own account: an occupant model or metric present in a cited primary study is insufficient unless the review discusses, tabulates, or classifies it. Where evidence falls between two adjacent levels, the lower score is recorded unless the higher criterion is clearly satisfied.

Worked boundary examples. The examples below are taken from the survey-level audit tables that follow.

Summary fields. Each survey-level audit table also carries three narrative fields: metric orientation, dominant occupant framing, and the main RQ4 consequence. These summarize the coded scores in prose and are not themselves coded values.

Table 6: Worked boundary examples for RQ4 coding.

Review	Dimension	Score	Basis for the boundary
Al-Ani and Das (2022)	Occupant framing, survey	1	Occupants appear mainly through comfort as an optimization objective, with little treatment as behavioral agents, feedback sources, or personalization targets. Boundary against 2.
Wang and Hong (2020)	Occupant framing, survey	2	Occupants are discussed through comfort, environmental adaptation, and feedback, but occupant behaviour and personalization do not structure the review. Boundary against both 1 and 3.
Han et al. (2021)	Occupant framing, survey	3	Occupants are represented across presence, activity, comfort feedback, state information, state-transition influence, and reward-related disturbance, which organizes the review.
Weinberg et al. (2023)	Metric visibility, survey	2	Energy, cost, and comfort violations are discussed as reward components, but human-centered metrics do not structure the review. Boundary against 3.
Han et al. (2021)	Metric imbalance, survey	2	The energy and comfort structure is clear, but the review is not mainly organized as a metric-imbalance review. Boundary against 3.
Kannari et al. (2025)	Metric imbalance, survey	3	The review shows explicitly that most real-world studies simplify comfort to indoor temperature while broader comfort variables remain less developed.
Al-Ani and Das (2022)	Gap signal, survey	1	Real applications and case-study coverage are noted as concerns, but occupant-centered gaps are not developed. Boundary against 2.